Hui Guo

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Education

– **Ph.D. in Computer Science, Western University** 09/2021–Present
Department of Computer Science
Advisors: Prof. Grace Y. Yi and Prof. Boyu Wang

– **B.Sc. in Mathematics and Applied Mathematics, Nankai University** 09/2016–06/2021
Thesis: A Generalization on Conditional Distortion Risk Measures

– **B.Econ. in Finance, Nankai University** 09/2016–06/2021
Thesis: Pairs Trading Strategy Based on Hurst Exponent

Refereed Publications and Manuscripts

1. **Hui Guo**, Boyu Wang, Grace Y. Yi (2026). Dual-Granularity Learning for Regression with Continuous Noisy Labels. ***Neural Information Processing Systems (NeurIPS)***.

2. **Hui Guo**, Grace Y. Yi, Boyu Wang (2026). Addressing both Variable Selection and Misclassified Responses with Parametric and Semiparametric Methods. ***Bernoulli*** 32(2): 1303–1327. [paper] [code] [arXiv]

3. Yuan Bian, Yu Shi, **Hui Guo**, Grace Y. Yi, Wenqing He (2025). Physician Effects in Critical Care: A Causal Inference Approach Through Propensity Weighting with Parametric and Super Learning Methods. ***Journal of Data Science*** 23(1): 130–148. [paper] [code]

4. Gezheng Xu*, **Hui Guo***, Li Yi, Charles Ling, Boyu Wang, Grace Y. Yi (2025). Revisiting Source-Free Domain Adaptation: A New Perspective via Uncertainty Control. ***International Conference on Learning Representations (ICLR)***. [paper] [code]
**Equal contribution*

5. **Hui Guo**, Grace Y. Yi, Boyu Wang (2024). Learning from Noisy Labels via Conditional Distributionally Robust Optimization. ***Neural Information Processing Systems (NeurIPS)***. [paper] [code] [arXiv]

6. **Hui Guo**, Boyu Wang, Grace Y. Yi (2023). Label Correction of Crowdsourced Noisy Annotations with an Instance-Dependent Noise Transition Model. ***Neural Information Processing Systems (NeurIPS)***. [paper] [code]

Awards and Distinctions

– **Robert and Ruth Lumsden Graduate Fellowship in Science, Western University** 2024–2025
Awarded annually to one PhD student in the Faculty of Science in recognition of exceptional academic achievement, research excellence, and leadership.

Selected Talks and Conference Presentations

– Learning with Noisy Labels: Variable Selection and Misclassification Probability Modeling
Lightning talk for Dr. Raymond Carroll's visit, Western University, June 2025.

– Revisiting Source-Free Domain Adaptation: A New Perspective via Uncertainty Control
International Conference on Learning Representations (ICLR), Singapore, April 2025.

– Learning from Noisy Labels via Conditional Distributionally Robust Optimization
Neural Information Processing Systems (NeurIPS), Vancouver, Canada, December 2024.

– Label Correction of Crowdsourced Noisy Annotations with an Instance-Dependent Noise Transition Model
Neural Information Processing Systems (NeurIPS), New Orleans, USA, December 2023.

– Variable Selection for Logistic Regression Models with Misclassified Response
Annual Meeting of the Statistical Society of Canada, Online, May 2022.

– Personalized Physician Recommendation for Critical Care Using the TreeSHAP Method (case study)
Annual Meeting of the Statistical Society of Canada, Online, May 2022.

Workshops & Events

– Remarkable, Vector Institute
Toronto, ON, February 2026

– Southern Ontario Learning Theory (SOLT) Workshop, Vector Institute
University of Waterloo, Waterloo, ON, November 2025

Academic Service

– **Conference reviewer:** ICML 2026; ICLR 2024–2026 (3 years); AISTATS 2024–2026 (3 years); NeurIPS 2024–2026 (3 years).

– **Journal reviewer:** *Foundations and Trends in Signal Processing*; *Neural Networks*.

Selected Teaching Experience

Teaching Assistant, Western University London, ON, Canada

– **CS2209: Applied Logic for Computer Science** 2021–2026 (7 terms)
An undergraduate core course for Computer Science students, with around 300 students, covering propositional and predicate logic, resolution, and logic programming in Prolog.

– **CS2035: Data Analysis and Visualization** 2022–2024 (3 terms)
An undergraduate elective for Computer Science students, with around 80 students, covering statistical data analysis and visualization techniques using MATLAB.

Technical Skills

– **Programming:** Python, R, MATLAB, C++, LaTeX, Prolog.